Chapter 12

Capstone: End-to-End ML Workflow

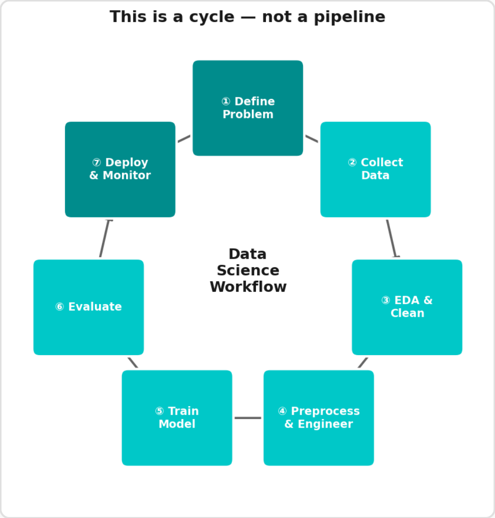
Session 4 | Final Chapter Putting it all together

Today’s Mission

April 15, 1912. RMS Titanic strikes an iceberg. 1,502 of 2,224 passengers die.

Can we predict who survived?

The Full ML Workflow



① Define → ② Explore → ③ Clean → ④ Engineer → ⑤ Train → ⑥ Evaluate → ⑦ Interpret → back to ①

The Dataset

Feature	Type	Notes
`Pclass`	ordinal	1st, 2nd, 3rd class
`Sex`	categorical	gender
`Age`	numeric	**many missing**
`SibSp`	numeric	siblings/spouses aboard
`Parch`	numeric	parents/children aboard
`Fare`	numeric	ticket price
`Embarked`	categorical	C, Q, S
`Cabin`	text	**mostly missing**

Target: `Survived` (0 = No, 1 = Yes)

What Does "Success" Mean?

- We want to **correctly identify survivors** → high recall
- We also want **predictions to be reliable** → high precision
- **F1-score** balances both

A False Negative here = predicted dead, actually survived

Step 1 — Explore

- `df.shape`, `df.dtypes`, `df.isnull().sum()`
- Survival rate by gender, class, age
- **The "women and children first" signal is in the data**

Step 2 — Clean

- `Age` → impute with median **per class** (not global median!)
- `Embarked` → fill with mode (only 2 missing)
- Drop: `Cabin` (too sparse), `Name`, `Ticket`, `PassengerId`
- Encode: `Sex` → 0/1, `Embarked` → one-hot

Step 3 — Feature Engineering

```
df['FamilySize'] = df['SibSp'] + df['Parch'] + 1
df['IsAlone'] = (df['FamilySize'] == 1).astype(int)
```

Why? Solo travelers had different survival odds **Domain knowledge + creativity = better features**

Step 4 — Train Multiple Models

Model	Idea
Logistic Regression	Linear boundary in probability space
Random Forest	Ensemble of decision trees
Gradient Boosting	Sequential error correction

Cross-validate all → compare F1 + accuracy

Step 5 — Interpret

- Which features matter most? (Random Forest importances)
- Confusion matrix: where does the model fail?
- What does Sex importance tell us about history?

Typical Results on Titanic

Model	Accuracy	F1-Score
Logistic Regression	~80%	~75%
Random Forest	~82%	~78%
Gradient Boosting	~83%	~79%

What this tells us:

- All models substantially beat the "always predict majority" baseline (~62%)
- Tree ensembles tend to win on tabular data with categorical features
- The gap between models is small → feature engineering matters more than algorithm choice
- Sex + Pclass are consistently the top features across all models

What We’ve Covered

Chapter	Topic
Ch01	Data Science workflow — **this is it**
Ch02	Cleaning and encoding — **most of the work**
Ch03–06	Fit / predict / evaluate / cross-validate
Ch07–09	Clustering and dimensionality reduction
Ch10–11	Reinforcement learning — a different paradigm

You Can Now

- Load and explore **any** dataset
- Clean and preprocess data correctly
- Train, evaluate, and compare ML models
- Understand clustering and dimensionality reduction
- Know what reinforcement learning is

What Comes Next

- **Deep Learning** — neural networks
- **Hyperparameter tuning** — GridSearchCV, Optuna
- **Model deployment** — serving predictions in production
- **MLOps** — monitoring, retraining, pipelines

Keep Going

- **Kaggle** — real competitions, real data
- **Fast.ai** — practical deep learning
- **"Hands-On ML"** — Aurélien Géron (the book)

Now — open the notebook

`03-exercises/ch12_capstone_exercises.ipynb`

35 minutes. Full workflow. You've got this.